

HANDY TOOLS For Programmers And Developers

The following software, besides other, can be found in the DVD accompanying EFY Plus edition of this magazine

OpenCV 3.4.1

OpenCV (Open Source Computer Vision Library) is a library of programs released under BSD licence. It is compatible with Windows, Linux, Mac OS as well as mobile platforms like Android and iOS. The library contains interfaces for JAVA, Python, C++ and MATLAB/OCTAVE programming. Expansion to other program wrappers like C#, Perl, Ch, Haskell and Ruby is possible as well. The suite is developed in C and C++, and can utilise multi-core processing. Its latest version includes bug fixes, improved DNN modules, over 250 patches and more.

Great Cow BASIC

It is an integrated development environment (IDE) to program microcontroller PIC units or Atmel AVR 8-bit microcontrollers. It can be used by beginners as well as professionals. The IDE is developed using Synwrite and is called GCB@Syn. It comes with an integrated compiler and supports a number of other tools including PPS for Microchip Peripheral Pin Select, TinyMultiBootLoader+, Xpress-Loader, IPE-Pusher, Pickett2 GUI, Pickett3 GUI, and Microchip Xpress board programmer and IPE automation tool. The latest version of GCB comes with auto-complete feature, function and sub-function support, syntax highlighting and code folding in editor, one-click assemble/compile/flash microcontroller and so on.

SWIG

Simplified Wrapper and Interface Generator (SWIG) lets you create a mesh between your programs created in C, C++ and other scripting suites. Any native function or code built in C/C++ can be called and used through other supporting scripting languages

including Perl, PHP, Python, R, Ruby, Lua, Tcl as well as programming languages like C#, Java, JavaScript, Go, Scilab, Octave, Modula-3 and so on. SWIG uses ANSI C/C++ declarations and supports all native data types and data structures including arrays, pointers and structures. It provides output in XML format or Lisp S-expressions, and it also creates documentation in HTML, LaTeX and ASCII formats.

HEXedit

HEXedit is a free, open-source binary editor with a simple interface and easy learning curve. Developed with C++, it lets users analyse the content and edit files of any size in decimal, hexadecimal and ASCII formats. It has features similar to Microsoft's Visual Studio and Word tools, including text highlighter, mark facility to go back to a portion in a file when needed, bookmarking, background search, clipboard, file import and export, jump tools, keystroke macros and so on.

Hibernate

Hibernate is an object relational mapping (ORM) suite for Java that allows users to map object-oriented programming models to relational databases. It does so primarily by mapping Java classes to database tables, and Java datatypes to SQL datatypes. Through data query and retrieval facilities, it automatically handles the otherwise manual process of object conversion from Java. It uses Hibernate Query Language (HQL)—a language similar in syntax with SQL written against Hibernate's data objects. The latest version supports Java8, Karaf, Jcache, JPA 2.2 and more.

TortoiseSVN

This is a free Apache Subversion client that can be used as a Windows Shell

extension. It is useful for programmers to maintain different versions of program codes simultaneously. Its intuitive interface does not require the Subversion command line to run, making it easy to use for all user levels. It also provides icon overlay, which displays the status of Subversion working copies. The latest version includes a major bug fix in the form of rectification of code execution through malicious svn + ssh URLs.

Moodle

Moodle is an open source free learning management software with highly customisable features. A large list of plugins for various purposes like knowledge, activity, communication, collaboration teaching and so on can be accepted by Moodle to build a stronger learning suite. Some examples include StudentQuiz, Turnitin's Moodle Direct V2 activity plugin, SAML2 Single sign-on, Content Pages, Feedback Plugin, Advanced Notifications, YU Kaltura Media Assignment, Media Resource and Media Gallery and Custom Course Menu.

Keepass Password Safe

This open source, free software is a Windows-major password management system. It also supports MacOS and Linux with the use of Mono. Users may need to manage different passwords for different accounts—personal email, work email and so on. Keepass can store all this information in various fields using username, password, free form notes, encrypted file attachments and so on. The latest version also features file import; export of stored data in TXT, HTML, XML or CSV format; multi-user support; and auto-type and global-autotype hot keys. **EFY**